- **L3 MIASHS/Ingémath/METIS**
- **Université Paris Cité**
- Année 2024-2025
- Course Homepage

- Moodle

⚠ Toutes les questions portent sur les schémas `pagila` et `world` rappelés ci-dessous.
Pour chaque question, proposer une requête écrite en algèbre relationnelle *OU* en `SQL`.

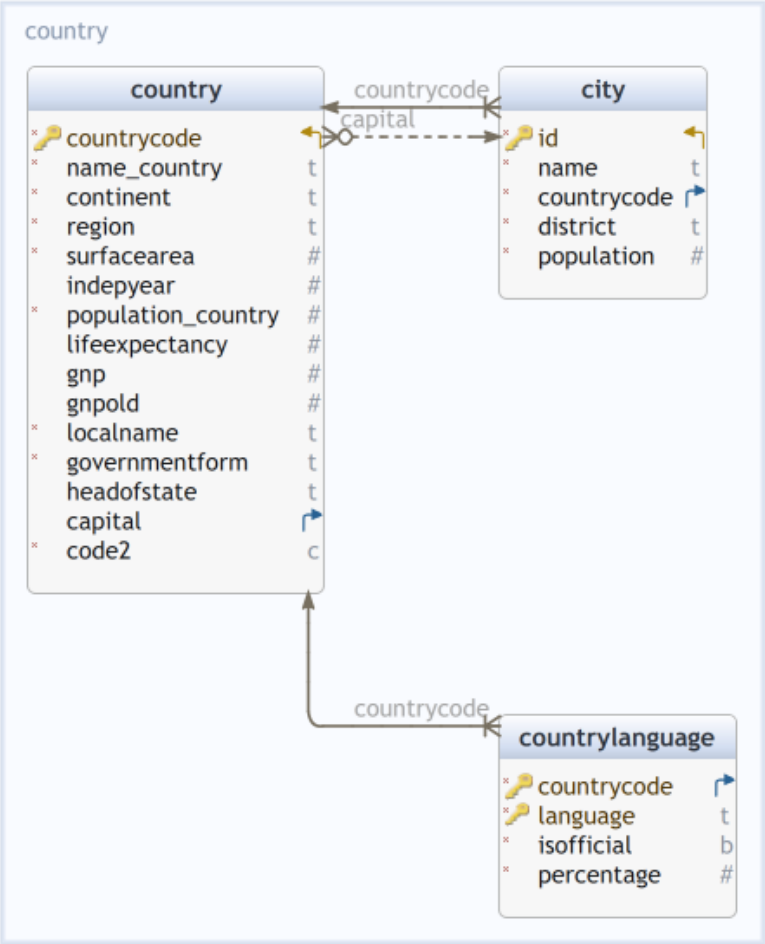


FIG. 1 : Schéma `world`

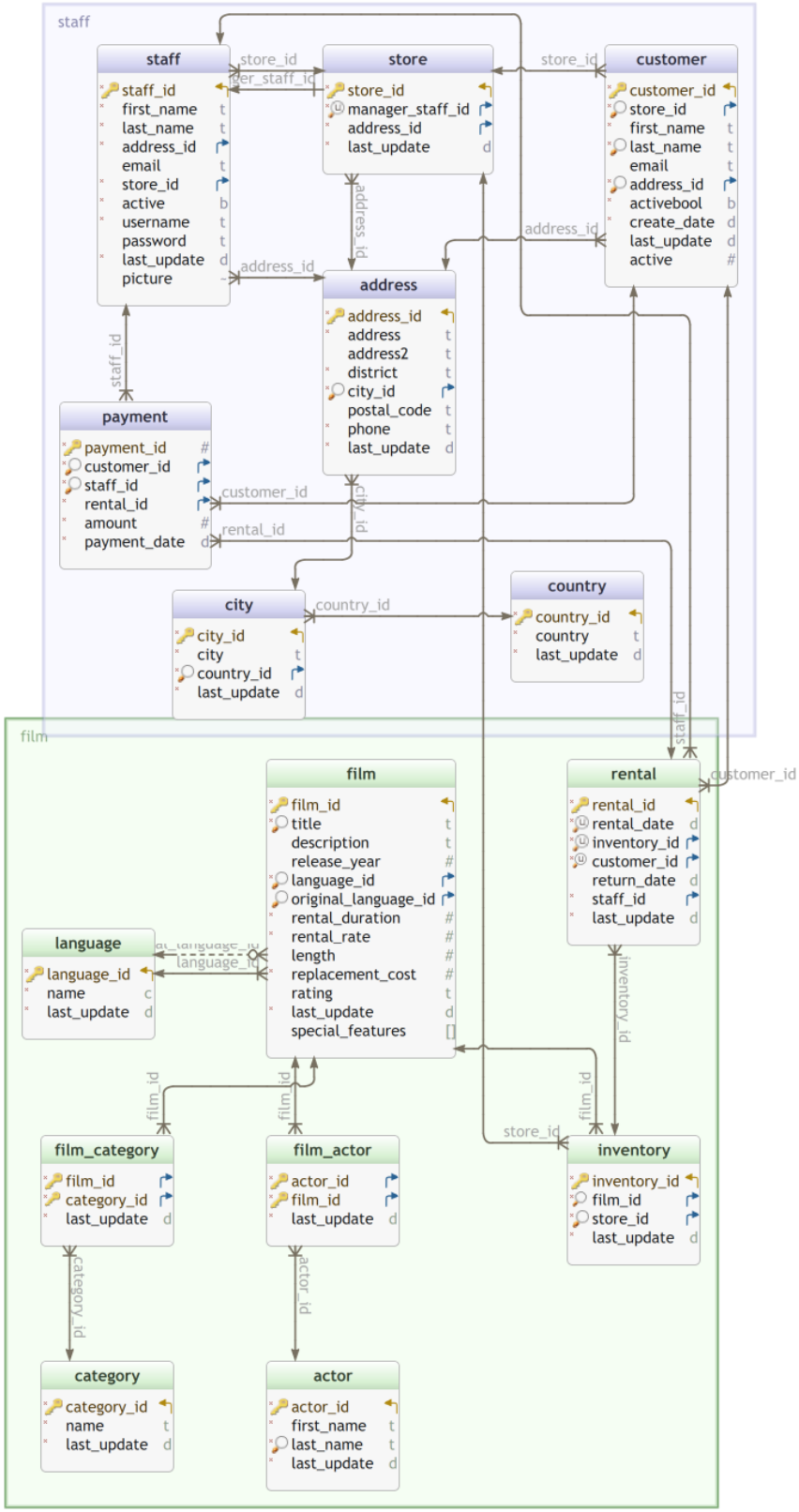


FIG. 2 : Schéma `pagila`, films

Schéma `pagila`

1. Lister la somme des paiments encaissés par année et par magasin.

En PostgreSQL, on obtient le (numéro du) mois à partir d'une valeur de type `timestamp` avec la fonction `EXTRACT()`. On peut extraire les autres éléments de l'estampille de façon semblable, par exemple :

```
SELECT
  EXTRACT(YEAR FROM rental_date) AS annee
FROM
  pagila.rental;
```

Solution

```
SELECT
  st.store_id,
  EXTRACT(YEAR FROM pay.payment_date) AS annee,
  SUM(amount) AS payments
FROM
  pagila.payment pay
JOIN
  pagila.staff st USING(staff_id)
GROUP BY
  st.store_id,
  EXTRACT(YEAR FROM pay.payment_date)
;
```

2. Lister l'identifiant, le nom, le prénom, de chaque employé ainsi que le total des paiements perçus par cet employé, et sa part dans le revenu de son magasin.

Solution

```
SELECT
  st.store_id,
  st.staff_id,
  SUM(pay.amount) AS encaisse
FROM
  pagila.staff st
JOIN
  pagila.payment pay USING(staff_id)
GROUP BY ROLLUP(st.store_id, st.staff_id) ;
```

3. Lister par catégorie de film, les cinq titres qui ont engendré le plus gros chiffre d'affaire/revenu (somme des paiements perçus)

Solution

```
WITH film_payment AS (
   SELECT
     film_id,
     fi.title,
     SUM(amount) AS caffaires
   FROM
     pagila.payment pay
   JOIN
     pagila.rental re USING(rental_id)
   JOIN
     pagila.inventory inv USING(inventory_id)
   JOIN
     pagila.film fi USING(film_id)
   GROUP BY
     film_id, fi.title
), rank_per_cat AS (
  SELECT
    category_id,
    ca.name,
    film_id,
    title,
    caffaires,
    RANK() OVER win AS rnk
  FROM
    film_payment fp
  JOIN
    pagila.film_category fc USING(film_id)
  JOIN
    pagila.category ca USING(category_id)
  WINDOW win AS (PARTITION BY category_id ORDER BY caffaires DESC)
)

SELECT
  name,
  title,
  caffaires,
  rnk
FROM rank_per_cat
WHERE rnk <= 5
ORDER BY name, rnk
;
```

Schéma `world`

4. Lister pour chaque pays, la proportion de la population qui utilise l'une des deux langues les plus populaires du pays,

Solution

```
WITH ranking AS (
   SELECT
     countrycode, name_country, language, percentage,
     RANK() OVER win AS rnk
   FROM
     world.country co
   JOIN
     world.countrylanguage cl USING(countrycode)
   WINDOW
     win AS (PARTITION by countrycode ORDER BY percentage DESC)
)

SELECT
  countrycode, name_country,
  SUM(percentage) AS prop
FROM
  ranking
WHERE
  rnk <= 2
GROUP BY countrycode, name_country
ORDER BY prop DESC;
```

Solution

5. Lister pour chaque région (attribut `region` de `country`) les 10 villes les plus peuplées.

Solution

```
WITH R AS (
   SELECT
     co.countrycode, co.name_country,
     ci.name,
     RANK() OVER win AS rnk
   FROM
     world.country co
   JOIN
     world.city ci USING(countrycode)
   WINDOW win AS (PARTITION BY co.countrycode ORDER BY ci.population DESC)
 )

SELECT
  name_country,
  string_agg(name, ', ')
FROM
  R
WHERE rnk <= 2
GROUP BY countrycode, name_country
 ;
```

6. Lister pour chaque langue, les deux pays où on trouve le plus grand nombre d'utilisateurs.

Solution

```
WITH R AS (
   SELECT
     language,
     name_country,
     percentage*population_country/100.0 AS pop_loc,
     RANK() OVER win AS rnk
   FROM
     world.countrylanguage cl
   NATURAL JOIN
     world.country
   WHERE
     population_country IS NOT NULL
   WINDOW win AS (PARTITION BY language ORDER BY percentage*population_country DESC)
)

SELECT
  language,
  string_agg(name_country, ', ') AS pays, SUM(pop_loc) AS loc
FROM
  R
WHERE rnk <=  2
GROUP BY language ;
```

7. Lister les pays en donnant leur rang par `gnp` (`pnb`) par habitant décroissant et leur rang par espérance de vie à la naissance (`lifeexpectancy`) décroissante.

Solution

```
SELECT
  name_country,
  lifeexpectancy ,
  RANK() OVER win_gnp AS rnk_gnppercap,
  RANK() OVER win_life AS rnk_lifeexp
FROM
  world.country
WHERE
  gnp IS NOT NULL AND
  population_country IS NOT NULL AND
  lifeexpectancy IS NOT NULL
WINDOW
  win_gnp AS (ORDER BY gnp/population_country DESC),
  win_life AS (ORDER BY lifeexpectancy DESC)
ORDER BY lifeexpectancy DESC ;
```